THE LAWS OF SOFTWARE TEAMS

The 90-90 rule

The last 10% of the code takes the other 90% of the time

swipe →

Ivan Rykov

90% done
Still 90% done

FEATURE · NEW CHECKOUT

WEEK 4

90%

done

no change for two weeks

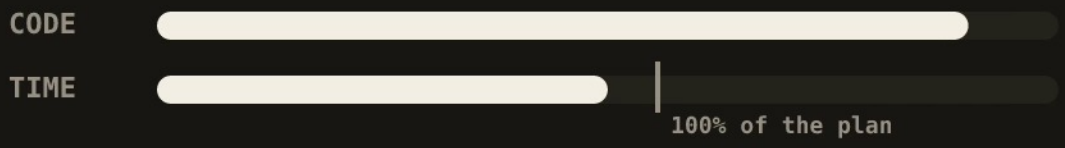
This pattern has a rule

illustration

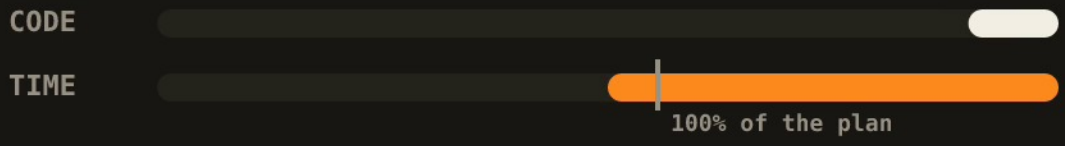
The 90-90 rule

Tom Cargill · Bell Labs

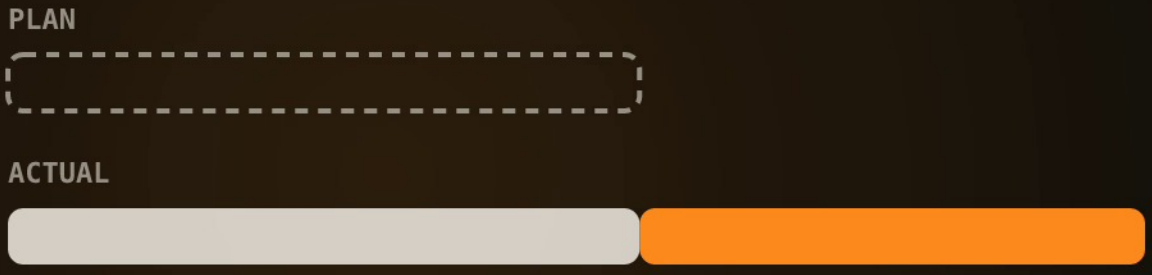
"The first 90 percent of the code accounts for the first 90 percent of the development time



The remaining 10 percent of the code accounts for the other 90 percent of the development time"



90% + 90% = 180%



JON BENTLEY · 1985

"Programming Pearls", Communications of the ACM: the "Rule of Credibility"

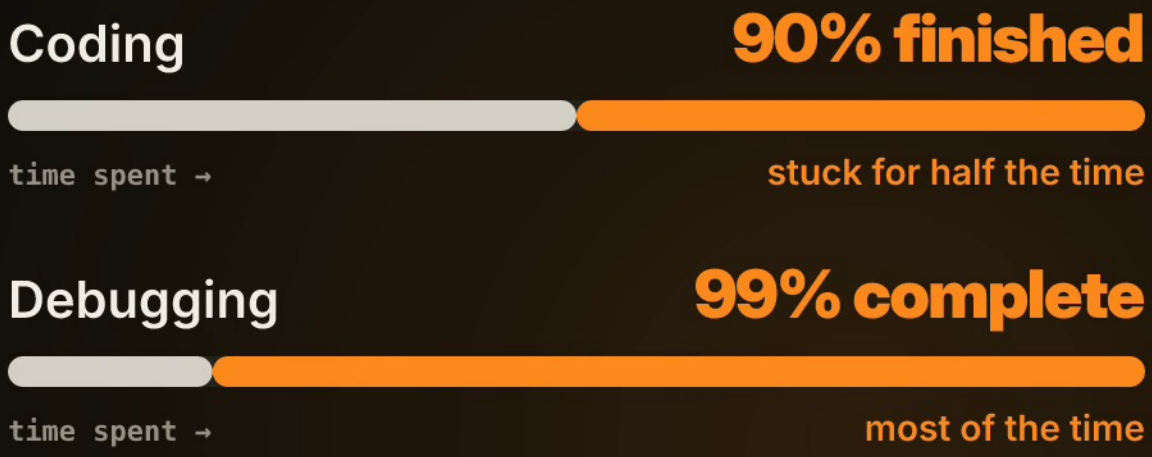
Teams still recognise it

Ivan Rykov

Brooks saw it in 1975

THE MYTHICAL MAN-MONTH · 1975

Fred Brooks · ch. 14, Hatching a Catastrophe



"Coding ... is '90 percent finished' for half of the total coding time. Debugging is '99 percent complete' most of the time"
bars illustrative

Bad news waits until the end

Two studies of large government projects



The estimate does not move
Then it moves all at once

King & Wilson, 1967, as quoted by Brooks (ch. 14) · chart illustrative

The last 10% is not the main code

main code

INSIDE THE LAST 10%

- edge cases
- integration
- data migration
- monitoring
- documentation
- rollout

typical examples

07 WHAT TO DO

Plan the finish from day one

1. **Plan the last 10% on day one**
   edge cases, migration and rollout go into the plan
2. **Show working slices**
   a demo, not a percentage
3. **Define done before you start**
   in production and used, not “code complete”

FRED BROOKS · 1975

“Milestones must be concrete, specific, measurable events, defined with knife-edge sharpness”

Done is not 90%

~~90% done~~

Done means:

- it is in production
- people use it

TAKEAWAY

Done is not 90%

Sources: Tom Cargill, via Jon Bentley, CACM, 1985 · Fred Brooks, The Mythical Man-Month, 1975

next: Zawinski's law